TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Estimation of Copper by the Iodometric Method

Copper weighed without a balance: Cu^{2+} sets iodine free mole for mole, the thiosulphate standardised in Experiment 5 counts the iodine, and a late spoonful of thiocyanate sharpens the finish.

CHE-1206 · CHEMISTRY-II SESSIONAL · EXPERIMENT 6 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the amount of copper in a supplied copper sulphate solution by iodometric titration with the standardised sodium thiosulphate of Experiment 5.

Copper(II) is an oxidising agent strong enough to take electrons from iodide. Add excess KI to the copper solution and two things happen at once: the copper falls as cream-white copper(I) iodide, and an exactly equivalent amount of iodine is set free —

$$2\ Cu^{2+} + 4\ I^- \rightarrow Cu_2I_2\downarrow + I_2$$

The liberated iodine (held dissolved by the excess iodide, as I_3^-) is then titrated with standard thiosulphate, starch being added near the end as in Experiment 5:

$$I_2 + 2\ S_2O_3^{2-} \rightarrow 2\ I^- + S_4O_6^{2-}$$

Count the chain: 2 mol Cu^{2+} free 1 mol I_2, and 1 mol I_2 consumes 2 mol $S_2O_3^{2-}$ — so **1 mol Cu^{2+} ≡ 1 mol $S_2O_3^{2-}$**, and since the copper changes by one electron ($Cu^{2+} \rightarrow Cu^+$), the equivalent weight of copper is simply its atomic weight, **63.55 g per equivalent** (and of $CuSO_4 \cdot 5H_2O$, 249.7 g per equivalent). One millilitre of 1 N thiosulphate therefore corresponds to 63.55 mg of copper.

The medium is part of the method. The titration is run in *weakly* acid solution — acetic acid. In strong mineral acid, air oxidises iodide rapidly (inventing iodine and inflating the result), and nitric acid, itself an oxidant of iodide, must be absent altogether. Two finishing touches protect the end point. First, the CuI precipitate **adsorbs iodine** on its surface, so the blue colour fades reluctantly. Second, near the end a little potassium thiocyanate converts the surface to CuSCN — $CuI + SCN^- \rightarrow CuSCN + I^-$ — which adsorbs far less; the trapped iodine is released and the blue vanishes sharply over a clean white precipitate. Thiocyanate is added *late*, never early: iodine oxidises thiocyanate itself, so an early addition would consume iodine and give a low result.

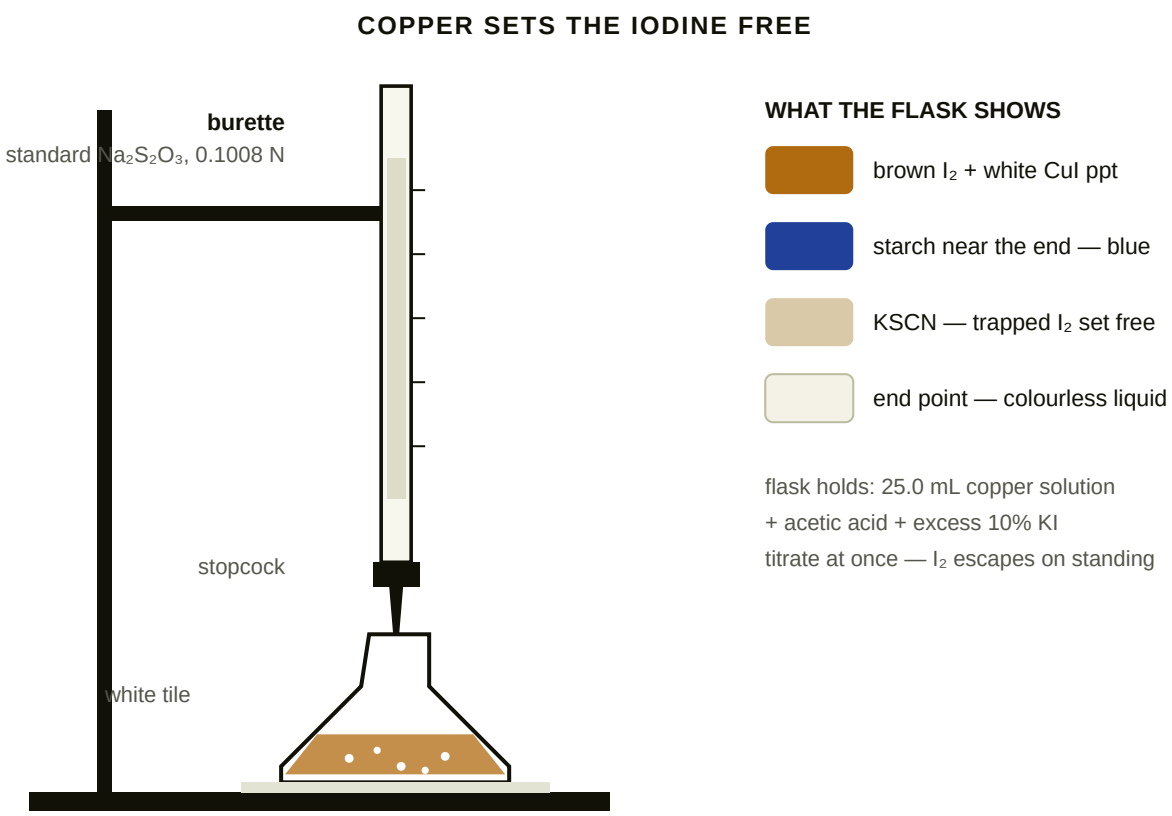


Figure 1 — The copper titration. Unlike Experiment 5 there is no coloured ion left behind: at the end point the liquid is colourless and only the white precipitate shows against the tile.

2. Apparatus and solutions

- Burette, 25 mL pipette, conical flasks, white tile, measuring cylinders.
- The supplied copper sulphate solution (unknown strength — in the model run below it was prepared from about 6.20 g of $CuSO_4 \cdot 5H_2O$ per 250 mL, the figure the analysis has to recover).
- Standard sodium thiosulphate, **0.1008 N** (standardised in Experiment 5 — this experiment is why that standardisation mattered).
- Potassium iodide solution, 10%; glacial acetic acid; starch solution, 1%, fresh; potassium thiocyanate solution, 10% (ammonium thiocyanate serves as well).

3. Procedure

1. Rinse and fill the burette with the standard thiosulphate (0.1008 N).
2. Pipette 25.0 mL of the copper solution into a conical flask. Add about 2 mL of glacial acetic acid (the medium should smell faintly of vinegar — weakly acid, no more), then 10 mL of 10% KI solution. Iodine is liberated at once: the mixture turns brown and a cream-white precipitate of CuI appears.
3. Titrate **without delay** with the thiosulphate, swirling well, until the brown fades to pale yellow.

4. Add 1–2 mL of starch (deep blue), and continue until the blue is nearly gone. Now add about 5 mL of 10% KSCN solution and shake vigorously: the precipitate turns clean white and any blue trapped on it is released — titrate the last of the blue away drop by drop.
5. The end point is reached when the blue colour **just disappears**, leaving a colourless liquid over the white precipitate, judged against the white tile. Record the titre and repeat to concordance.

4. Worked calculation (model data)

COPPER RESULT

Aliquot taken: 25.0 mL of the copper solution. Titrant: $Na_2S_2O_3$, 0.1008 N. Concordant titre: 24.6 mL.

Milliequivalents of thiosulphate used = 0.1008 × 24.6 = 2.480 meq — and equivalents of titrant = equivalents of copper (1 Cu^{2+} ≡ 1 $S_2O_3^{2-}$).

Copper in the 25.0 mL aliquot = 2.480 meq × 63.55 mg meq^{-1} = **157.6 mg**.

Copper in the full 250 mL = 157.6 × 10 = **1.576 g** (i.e. 6.30 g of Cu per litre).

Expressed as the salt weighed out: 1.576 g Cu × (249.7/63.55) = **6.19 g of $CuSO_4 \cdot 5H_2O$ per 250 mL** — against 6.20 g actually dissolved, a recovery of about 99.9%. The iodometric chain, anchored in Experiment 5 to a weighed primary standard, has weighed the copper to within a centigram without the copper ever touching a balance during the titration.

5. Observation table (as recorded)

TRIAL	INITIAL READING (ML)	FINAL READING (ML)	TITRE (ML)
1 (rough)	0.0	24.9	24.9
2	0.0	24.6	24.6
3	0.0	24.6	24.6

Concordant titre: 24.6 mL. Watch the rough titre: it ran high because the end point was approached without the thiocyanate release — the adsorbed iodine on the CuI kept the blue alive and invited overshoot. That is the whole lesson of the KSCN step, visible in one row of the table.

6. Precautions

- **Add KSCN only near the end point.** Iodine oxidises thiocyanate, so an early addition quietly consumes iodine that the burette should have counted — a low result. Added late, it only exchanges

the precipitate's surface (CuI → CuSCN) and frees the adsorbed iodine for a sharp finish.

- Keep the medium **weakly acid (acetic acid)**. Strong mineral acid speeds air-oxidation of iodide, and any nitric acid present oxidises iodide directly — both invent iodine and inflate the copper figure.
- Use a genuine **excess of KI**: it drives the reduction of Cu^{2+} to completion and holds the liberated iodine in solution as I_3^-, cutting the loss of volatile I_2.
- Titrate at once after adding KI and swirl vigorously near the end — iodine sitting on precipitate surfaces only rejoins the titration when the flask is shaken.
- Starch late, as in Experiment 5, and judge the end point against a white tile: the last trace of blue over a white precipitate is easy to miss against a dark bench.
- Do not let the titrated flask stand and then "correct" it: air-oxidation in the acid medium slowly regenerates iodine and the blue creeps back (compare Experiment 5) — the first clean disappearance is the end point.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is this titration done in acetic acid rather than in dilute sulphuric acid?
2. The potassium iodide in this experiment does two distinct jobs. Name both.
3. Why is potassium thiocyanate added only near the end point, and what exactly does it do?
4. 25.0 mL of a copper solution requires 22.5 mL of 0.1000 N thiosulphate. Calculate the copper in the aliquot, and the copper content in g L^{-1}.
5. What colour change does the precipitate undergo during the titration, and why does the dirty colour matter?
6. One mole of $CuSO_4 \cdot 5H_2O$ ultimately consumes how many moles of thiosulphate? Trace the chain.

Answer key

1. The Cu^{2+}/I^- reaction is quantitative in weakly acid solution, while strong acid accelerates the air-oxidation of iodide ($4\ I^- + O_2 + 4\ H^+ \rightarrow 2\ I_2 + 2\ H_2O$), which liberates iodine not owed to copper and gives a high result. Acetic acid holds the pH in the narrow range where the reduction is complete but air-oxidation is slow. (Nitric acid is worse still — it oxidises iodide itself and must be absent.)
2. (i) It is the reducing agent that converts Cu^{2+} to Cu^+, being itself oxidised to the iodine that is titrated. (ii) In excess it dissolves and holds that iodine as I_3^-, preventing its loss by volatilisation before the burette can count it.

3. Thiocyanate converts the precipitate surface from CuI to CuSCN ($CuI + SCN^- \rightarrow CuSCN + I^-$). CuI adsorbs iodine strongly and releases it slowly, so the blue fades reluctantly; CuSCN adsorbs little, so the trapped iodine is freed and the end point is sharp. It is added late because iodine itself oxidises thiocyanate — added early, it would destroy some of the iodine being measured and give a low result.
4. Equivalents used = 0.1000 × 22.5 = 2.250 meq. Copper = 2.250 × 63.55 = **143.0 mg** in 25.0 mL. Per litre: 143.0 × (1000/25) = 5720 mg = **5.72 g L^{-1}**.
5. The precipitate forms cream-white (CuI) but looks dirty brown because iodine is adsorbed on its surface; after KSCN is added it becomes clean white (CuSCN). The dirty colour matters because that adsorbed iodine has not yet been titrated — it is analyte hiding on a solid, and until it is released the blue cannot disappear honestly.
6. One mole. Chain: $2\ Cu^{2+} + 4\ I^- \rightarrow Cu_2I_2 + I_2$ frees half a mole of I_2 per mole of copper, and $I_2 + 2\ S_2O_3^{2-}$ consumes two thiosulphates per I_2 — ½ × 2 = 1 mol of thiosulphate per mole of $CuSO_4 \cdot 5H_2O$.
